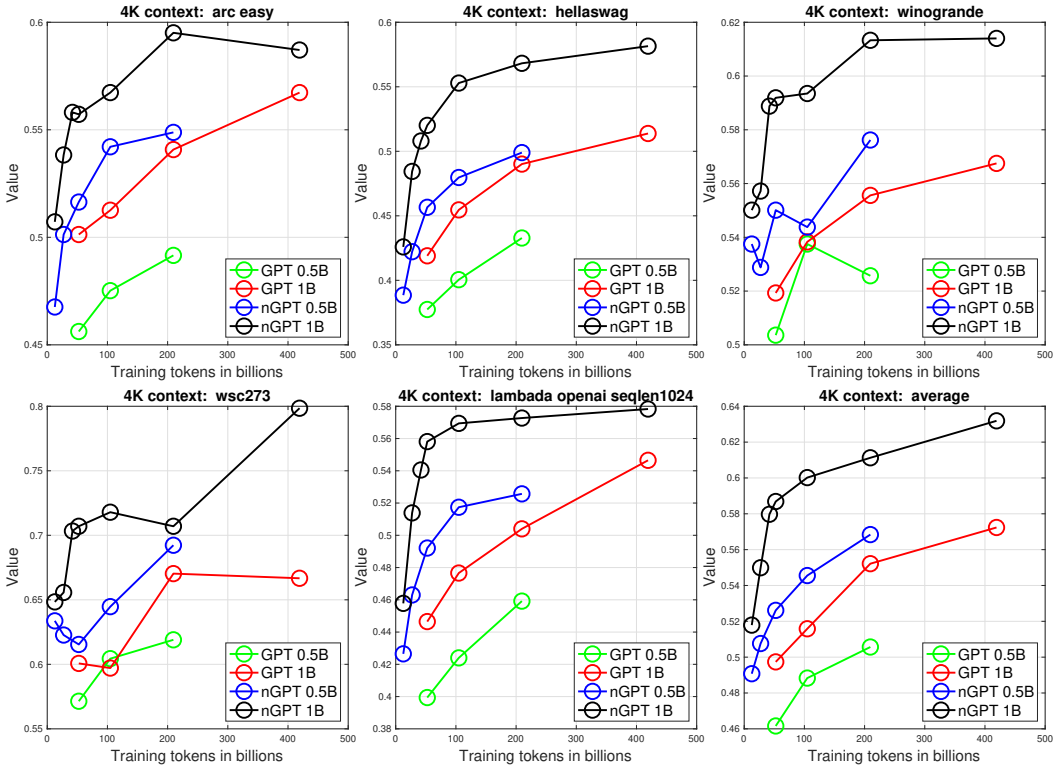

Figure 3: Models trained with 4k context length. Final performance (**y-axis**) on a set of downstream tasks and their average value (**Bottom-Right**) for different computation budgets in tokens (**x-axis**).

Figure 2 illustrates how the performance gap between nGPT and GPT scales across three axes: total token budget, context length, and network size. Training the 0.5B and 1B nGPT models is approximately 4x, 10x, and 20x faster at context lengths of 1k, 4k, and 8k tokens, respectively.

Figure 3 shows a similar pattern across downstream tasks, confirming that the acceleration is not only reflected in perplexity but also in task performance. Figures 8 and 10 in the Appendix provide results for 1k and 8k context lengths. We observe some saturation for the longest runs of nGPT, suggesting that the model capacity is nearly reached for this number of trainable model parameters.

3.2 Inspection of network parameters

Figure 4 shows that, while nGPT maintains a fixed norm for embeddings (by design), GPT exhibits significant variation. The distribution of eigenvalues, computed from the covariance matrix of embeddings and normalized by their median, reveals that GPT's input embeddings have a higher condition number, especially in the 1B model. The distribution of pairwise dot products between embeddings indicates that even in nGPT, embeddings are not uniformly distributed across the hypersphere (where the dot product would approach 0), but instead form clusters—possibly reflecting natural patterns in language data. Dot products in GPT tend to have higher values due to its embeddings forming a hyper-ellipsoid, as suggested by the spread of vector norms. The ill-conditioned nature of GPT's input embeddings could lead to computational issues involving these embeddings.

Figure 5 shows the median condition numbers (across heads) for attention and MLP matrices at different layer depths—24 layers for the 0.5B model and 36 layers for the 1B model. GPT models exhibit significantly higher condition numbers in their attention matrices compared to nGPT. A closer inspection of these matrices (the 3rd and 4th layers are in Figure 12 and Figure 13 of Appendix) suggests that they degenerate into lower-rank matrices, potentially reducing the learning capacity of these blocks. One could argue that the elevated condition numbers are influenced by the norms of the vectors in these matrices. Our post-training normalization of these matrices is depicted

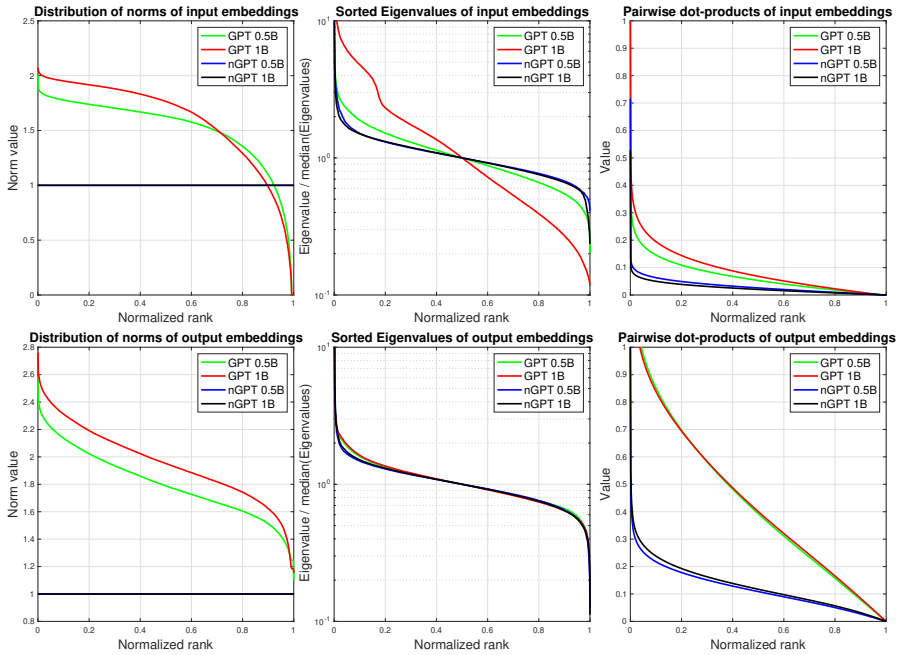

Figure 4: **Left**: Distribution of norms of vectors from input (**Top line**) and output (**Bottom line**) embedding matrices. **Middle**: Distribution of eigenvalues divided by its median value. **Right**: Pairwise distribution of dot products between embeddings. Models are trained for 100k iterations.

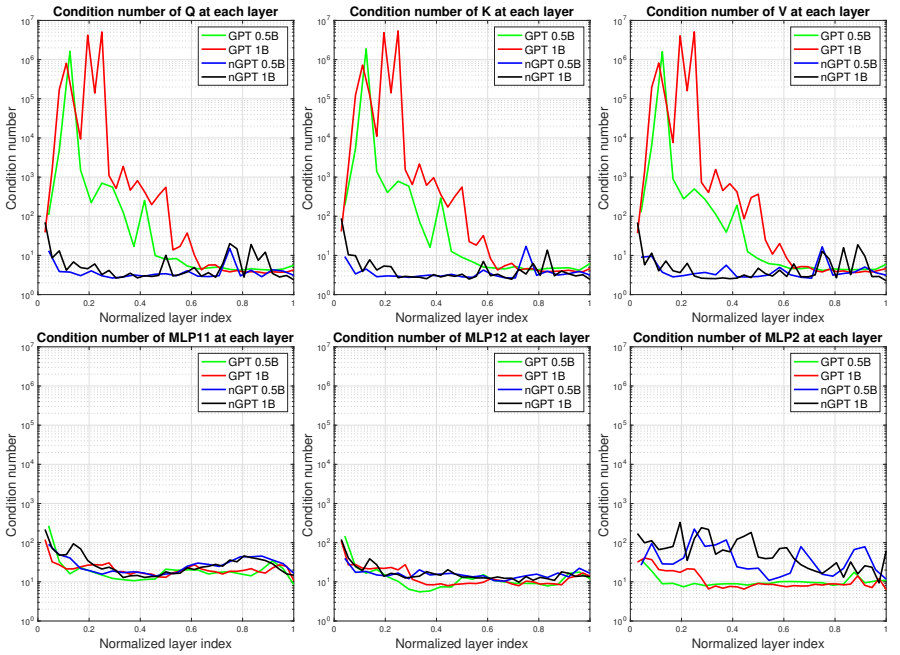

Figure 5: Median condition numbers for attention and MLP matrices at different layer depth (24 and 36 layers for 0.5B and 1B models, respectively). Models are trained for 100k iterations.

by the dotted lines in Figure 11 of the Appendix. While the adjusted condition numbers are reduced, they remain higher than those for nGPT, indicating potential rank deficiency. The need for such normalization highlights one of the issues that nGPT is specifically designed to address.

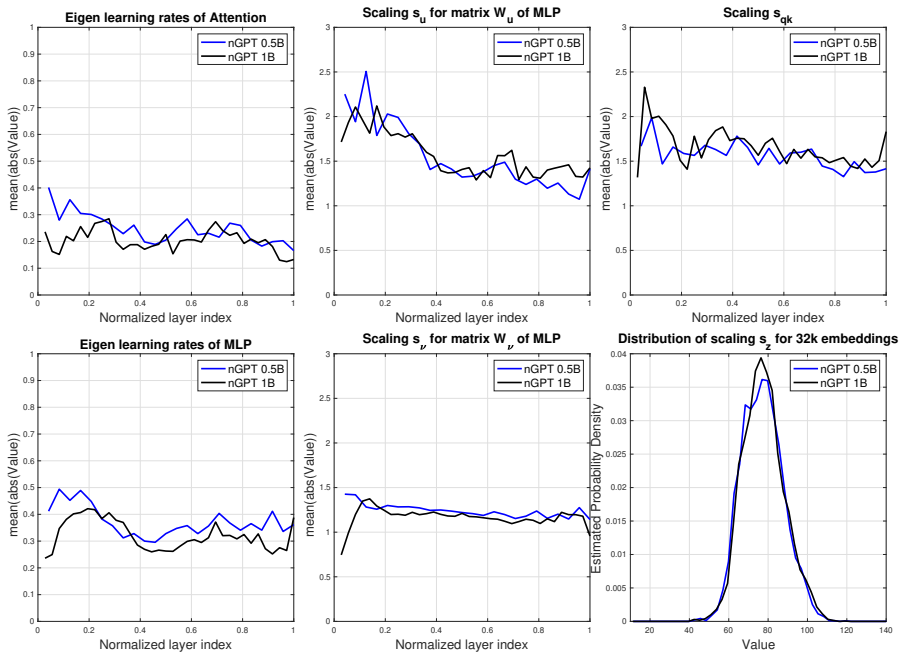

Figure 6: (**Left**): Eigen learning rates of Attention and MLP blocks. (**Middle**): Scaling factors applied to the intermediate states of MLP. (**Right**): Scaling factors applied before the QK dot product; distribution of per-vector scalings applied to logits. Models are trained for 100k iterations.

An important contribution of this work is the decoupling of predictions made by the Attention and MLP blocks from their impact on the hidden state $\boldsymbol{h}$. These contributions are controlled by the eigen learning rates $\boldsymbol{\alpha}_{\mathrm{A}}$ and $\boldsymbol{\alpha}_{\mathrm{M}}$. Their interpretation is straightforward: if $\boldsymbol{\alpha}_{\mathrm{A},i}$ for an embedding dimension $i \in \mathbb{R}^{d_{\text{model}}}$ is 0.2, then the update follows $h_i \leftarrow (1-0.2)h_i + 0.2h_{\mathrm{A},i}$. Thus, they directly quantify the contribution of $h_{\mathrm{A},i}$ into $\boldsymbol{h}$. Figure 6 shows the average absolute values of $\boldsymbol{h}_{\mathrm{A}}$ and $\boldsymbol{h}_{\mathrm{M}}$ at each layer. Notably, the network learns to take only modest steps (20%-30%) in the direction suggested by $\boldsymbol{h}_{\mathrm{A}}$ and $\boldsymbol{h}_{\mathrm{M}}$. The average magnitude of $\boldsymbol{\alpha}_{\mathrm{A}}$ decreases from 0.25 in the 0.5B network (24 layers) to 0.20 in the 1B network (36 layers). Meanwhile, $\boldsymbol{\alpha}_{\mathrm{M}}$ decreases from 0.37 to 0.32. If the parameter count per block is linked to its maximum predictive capacity (in our setup, MLP blocks have more parameters than Attention blocks), then the greater eigen learning values observed for MLPs could potentially be attributed to the higher quality predictions made by these blocks.

The scaling factors $\boldsymbol{s}_u$, $\boldsymbol{s}_\nu$ and $\boldsymbol{s}_{qk}$ remain relatively stable across layers. The value of $\boldsymbol{s}_\nu$ can be interpreted as a measure of the non-linearity of the SiLU function, which behaves like ReLU for large $\boldsymbol{s}_\nu$ and approximates a linear unit for values near 0 (see also Appendix A.1). The distribution of $\boldsymbol{s}_z$ is primarily characterized by its mean, which influences the temperature of the softmax during cross-entropy calculations. The introduced scaling factors $\boldsymbol{s}_{qk}, \boldsymbol{s}_u, \boldsymbol{s}_\nu$ and $\boldsymbol{s}_z$ seem to compensate for the removal of magnitude information when normalizing matrices and embeddings.

3.3 Ablation studies

Appendix A.9 summarizes numerous ablation experiments. An important finding is that having fixed (non-learnable) values for $\boldsymbol{s}_{qk}, \boldsymbol{s}_u, \boldsymbol{s}_\nu$ and a single global learnable value for $\boldsymbol{s}_z$ leads to only a slight degradation in accuracy. Therefore, our presented general case can be simplified and become easier to interpret. Appendix A.8 demonstrates that nGPT can handle longer contexts without requiring any modifications to RoPE.